

Section 3.6 Logarithmic Differentiation

$$\log (A \cdot B) = \log A + \log B$$

$$\log \left(\frac{A}{B} \right) = \log A - \log B$$

$$\log A^n = n \cdot \log A$$

$$y = \ln(A)$$

$$y' = \frac{1}{A} \cdot A'$$

$$y' = \frac{A'}{A}$$

Ex ① $y = \ln \left(\frac{x^2+7x}{2x+5} \right)$

$$y = \ln(x^2+7x) - \ln(2x+5)$$

$$y' = \frac{2x+7}{x^2+7x} - \frac{2}{2x+5}$$

Ex ② $y = x^x$

$$\ln y = \ln x^x$$

$$\ln y = x \cdot \ln x$$

$$\frac{y'}{y} = 1 \cdot \ln x + x \cdot \left(\frac{1}{x} \right)$$

$$y' = y (\ln x + 1)$$

$$y' = x^x (\ln x + 1)$$

$$\underline{\text{Ex}} \text{ (3)} \quad y = (5x+3)^{13} \cdot (x^2+5)^{-9} \cdot (2x+1)^2$$

$$\ln y = \ln (5x+3)^{13} + \ln (x^2+5)^{-9} + \ln (2x+1)^2$$

$$\ln y = 13 \cdot \ln (5x+3) + -9 \cdot \ln (x^2+5) + 2 \cdot \ln (2x+1)$$

$$\frac{y'}{y} = 13 \cdot \left(\frac{5}{5x+3} \right) - 9 \cdot \left(\frac{2x}{x^2+5} \right) + 2 \cdot \left(\frac{2}{2x+1} \right)$$

$$y' = y \left[\frac{65}{5x+3} - \frac{18x}{x^2+5} + \frac{4}{2x+1} \right]$$

$$y' = \frac{(5x+3)^{13} (2x+1)^2}{(x^2+5)^9} \left[\frac{65}{5x+3} - \frac{18x}{x^2+5} + \frac{4}{2x+1} \right]$$

Ex (4) Find y' in two ways when $x=1$.

$$y = \ln(\sqrt{x})$$

$$A) \quad y' = \frac{\frac{1}{2\sqrt{x}} \cdot 2\sqrt{x}}{\sqrt{x} \cdot 2\sqrt{x}}$$

$$y' = \frac{1}{2x}$$

$$y' = \frac{1}{2}$$

$$B) \quad y = \ln(\sqrt{x})$$

$$y = \ln(x^{1/2})$$

$$y = \frac{1}{2} \cdot \ln x$$

$$y' = \frac{1}{2} \cdot \frac{1}{x}$$

$$y' = \frac{1}{2x}$$

$$y' = \frac{1}{2}$$